

Problem Statement

Task: signal $X \in \mathbb{R}^{C \times T} \rightarrow$ Action Prediction

Dataset

Physionet $\Rightarrow$ 109 subjects, 64 channels
 1 subject $\Rightarrow$ 14 runs $\Rightarrow$ same session (real/imagery)
 2 runs $\rightarrow$ eyes open / closed (baseline)
 4 runs $\rightarrow$ Task 1 (real $\Rightarrow$ left vs right fist)
 Task 2 (imagined $\Rightarrow$ left vs right fist)
 Task 3 (real $\Rightarrow$ both fists vs both feet)
 Task 4 (imagined $\Rightarrow$ both fists vs both feet)

$\times 2$ more times $\Rightarrow$ Total 14 runs.

80 total 6 real vs 6 imagery runs.

1 run = 2 min 58 $\Rightarrow$ 125s

each run $\Rightarrow$ 4s interval between actions

4s rest (T_0), 4s action (T_1/T_2)

Total 15 actions (7-8 T_1/T_2)

Sampling rate $\Rightarrow$ 160 Hz $\rightarrow$ 125s $\Rightarrow$ 20000 samples

$C \in \{1, 2, 3, \dots, 64\}$

$T \in \{1, 2, \dots, 20000\}$

BCIIV-2a: 9 subjects, 22 EEG + 3 EOG

1 subject: 2 sessions (T & E) $\Rightarrow$ cross session

Actions: Left hand, right hand, both feet, Tongue

each session: 288 trials $\Rightarrow$ 4x4 actions

each trial: 7-8s $\Rightarrow$ 2s cue, 4s action, rest

Sampling frequency = 250 Hz.

Total time / session = 288 x (7-8) $\approx$ 2000-2400s

No. of samples: 250 x (2000-2400) = 500000 - 600000 samples

Dataset representation

$X_{i,a} \in \mathbb{R}^{C \times T}$; $i = 1, 2, 3, \dots, 109$ subjects, $a \in$ Action
 T' = epoch time (4s in Physionet, 3s/4s after cue)

Preprocessing

RIEMANNIAN MANIFOLD

Distance $\Leftrightarrow$ geodesic distance

$X_{i,a} \in \mathbb{R}^{C \times T}$; $C = \{1, 2, \dots, 64\}$, $T = \{1, 2, \dots, 20000\}$

$C_{i,a} = \frac{1}{C-1} X_{i,a} X_{i,a}^T \in \text{SPD}(C) \Rightarrow C \times C$ matrices

AIRM geodesic distance

$$d(C_1, C_2) = \|\log C C_1^{-1/2} C_2 C_1^{-1/2}\|_F$$

symmetric: $d(C_1, C_2) = d(C_2, C_1)$ — (i)

Mean (N_T : No of trials)

Riemannian Mean: $C_{R, \text{ref}, i} = \arg \min_C \sum_{e,a} d_R^2(C, C_{i,a,e})$
 of subject i

Euclidean Mean: $C_{E, \text{ref}, i} = \frac{1}{N_T} \sum_{e,a} C_{i,a,e}$ — (ii)

log-Euclidean Mean: $C_{LE, \text{ref}, i} = \frac{1}{N_T} \sum_{e,a} \log C_{i,a,e}$ — (iii)

Whitening

$C_{R, \text{ref}, i}$ is Riemannian mean of subject i .

Consider a matrix $C_{i,a,e}$ of subject i

$$C_{i,a,e}^{-1/2} C_{R, \text{ref}, i} C_{i,a,e}^{-1/2} \quad \text{--- (iv)}$$

$\Rightarrow$ We do this because, each subject i has $C_{R, \text{ref}, i}$ reference that is used to identify the subject. Therefore if C_{ref} of all subjects are I , then subjects are generalized & no cues for identifying subject

$$X'_{i,a,e} = C_{R, \text{ref}, i}^{-1/2} X_{i,a,e} \quad \text{--- (v)}$$

Here X' is normalized, $\sim x/\sigma$.

So $\mu/\sigma \Rightarrow$ deviation from I where all $x_i/\sigma \Rightarrow$ same μ . So this can be used specifically for classification.

RIEMANNIAN METHODS

Tangent Space Logistic Regression (CSP)

1. $X_{i,a,e}$, $a \in A$, $A \in \{0, 1, 2, 3\} \Rightarrow$ Physionet
 $e \in \{1, 2, \dots, N_T\}$

2. Compute Covariance $C_{i,a,e}$ of each $X_{i,a}$

$$C_{i,a,e} = \frac{1}{C-1} X_{i,a,e} X_{i,a,e}^T$$

3. Compute per subject C_{ref} .

Riemannian Mean: $C_{R, \text{ref}, i} = \arg \min_C \sum_e d_R^2(C, C_{i,e})$

Euclidean Mean: $C_{E, \text{ref}, i} = \frac{1}{N_T} \sum_e C_{i,e}$

Log-Euclidean Mean: $C_{LE, \text{ref}, i} = \frac{1}{N_T} \sum_e \log C_{i,e}$

4. Align the Covariances to the per subject ref such that C_{ref} of updated covariances per subject = I

We can use $C_{\text{ref}, i} = C_{R, \text{ref}, i}$, $C_{E, \text{ref}, i}$, $C_{LE, \text{ref}, i}$ depending on type of Alignment.

$$C'_{i,a,e} = C_{\text{ref}, i}^{-1/2} C_{i,a,e} C_{\text{ref}, i}^{-1/2}$$

5. Compute $C_{\text{ref}}^{\text{global}} = \arg \min_C \sum_{i,a,e} \|\log C^{-1/2} C'_{i,a,e} C^{-1/2}\|_F$

6. Project C' onto tangent space at $C_{g, \text{ref}}$

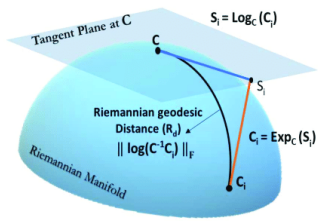
$$T_{i,a,e} = C_{g, \text{ref}}^{1/2} \log C C_{g, \text{ref}}^{-1/2} C'_{i,a,e} C_{g, \text{ref}}^{-1/2} C_{g, \text{ref}}^{1/2}$$

7. Perform Multinomial Logistic Regression.

8. $x_{i,a,e} = \text{vec}(T_{i,a,e})$

$$x_{(i-1) \times N + j} = \begin{cases} T_{ij} & i = j \\ \sqrt{2} T_{ij} & i \neq j \end{cases}$$

10. $z_k = w_k^T x_i + b_k$
 $\hat{y}_k = \sigma(z_k) = \frac{e^{z_k}}{\sum_k e^{z_k}}$ $k = 0, 1, 2, 3, 4$
 $\hat{y}_k = 1, 0, 1, 2, 3, 4$



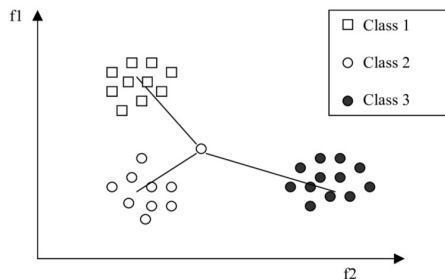
Minimum Distance to Mean (MDM)

Algorithm

1. $X_{i,a,e}$; $i \in \{1, 2, \dots, 109 \text{ subjects} \Rightarrow \text{Physionet}\}$
 $a \in \{0, 1, 2, 3\} \Rightarrow \text{actions}$
 $e \in \{1, 2, 3, \dots, N_T\}$
2. Compute Covariances $C_{i,a,e}$

$$C_{i,a,e} = \frac{1}{e-1} X_{i,a,e} X_{i,a,e}^T$$
3. Compute C_{ref} (global mean) for each class $K \in \{0, 1, 2, 3\} \Rightarrow 4 \text{ actions}$
 - * Euclidean Mean(K) = $\frac{1}{N} \sum_{i,e} C_{i,e,a} |_{a=K}$
 - * Riemannian Mean(K) = $\arg \min_C \sum_{i,e} d_R^2(C, C_{i,e,a}) |_{a=K}$ (MDRM)
 - * Log-Euclidean Mean(K) = $\frac{1}{N} \sum_{i,e} \log C_{i,e,a} |_{a=K}$
4. For Covariance $C_{i,e,a}$, for $K \in \{0, 1, 2, 3\}$

$$D_K(C_{i,e,a}, C_{ref,K}) = \|\log(C_{i,e,a}^{-1/2} C_{ref,K} C_{i,e,a}^{-1/2})\|_F$$
5. $\hat{y}_K = \arg \min_K D_K(C_{i,e,a}, C_{ref,K})$



Common Spatial Patterns (CSP)

Similar to PCA, maximizes variance, but CSP maximizes variance of 1 class while reducing variance of other classes. Finds all such vectors.

Algorithm

1. Consider K classes of signals $X_{K1}, X_{K2}, \dots, X_{Kn}$
2. Compute Covariances of each signal

$$C_{ki} = \frac{X_{ki} X_{ki}^T}{\text{trace}(X_{ki} X_{ki}^T)} \Rightarrow \text{trace normalization}$$
3. Compute average of covariances per class

$$C_K = \frac{1}{N_K} \sum_i C_{ki}$$
4. Now we have C_1, C_2, C_3, C_4 for 4 classes
5. So to maximize variance, maximize $J(w)$

$$J(w) = \frac{w^T C_K w}{w^T C_{-K} w} \approx \frac{w^T C_K w}{w^T C_T w}$$

where $C_K = \frac{1}{N_K} \sum_{i \in K} C_i$, $C_T = \sum_i C_i$
6. Find eigenvectors of $C_K w = \lambda (C_T) w$ (vii)
7. Whiten $\Rightarrow S_K = C_T^{-1/2} C_K C_T^{-1/2}$, Solve $S_K w = \lambda w$
8. Take top m & bottom m eigenvectors (Sort λ)

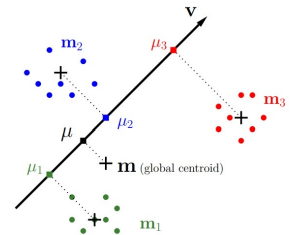
$$W_{MCSP} = [w_1, w_2, \dots, w_K]$$

$$X'_{ki} = W_{MCSP}^T X_{ki}$$
9. Use LDA or Multinomial Logistic regression
10. Concatenate X'_{ki} for all i & $K \in \{0, 1, 2, 3\}$
11. Logistic Regression $Z_K = w_K x'_{Ki} + b_K$

$$\hat{y}_K = \arg \max_c \frac{e^{Z_K}}{\sum e^{Z_K}} \text{ logits}$$

Linear Discriminant Analysis (LDA)

$\mu_j \Rightarrow$ projection
 $m_j \Rightarrow$ mean



LDA is Supervised method to find vectors/directions that maximizes the variance across classes & minimizes the variance within the class. So it's technically maximizes between-class variance & within-class variance.

$$J(v) = \frac{\sum_j (\mu_j - \bar{\mu})^2}{\sum_j s_j^2} \Rightarrow \frac{v^T S_b v}{v^T S_w v}$$

So $\mu_j, \bar{\mu}, s_j$ are all variance across points μ_j, m, x_j projected on vector v .

$$\mu_j = v^T m_j \quad \bar{\mu} = v^T m$$

$$S_b = \sum_j (\mu_j - \bar{\mu})(\mu_j - \bar{\mu})^T \Rightarrow \text{between class}$$

$$S_w = \sum_j \sum_{x \in c_j} (x - m_j)(x - m_j)^T \Rightarrow \text{within class}$$

Theorem: Suppose S_w is non-singular, then solution to maximize $J(v)$ is given by highest eigenvectors of $S_w^{-1} S_b \Rightarrow S_w^{-1} S_b v = \lambda v$

1. Compute S_w & S_b for the points X .
2. Solve the generalized eigenvalue problem $S_b v = \lambda S_w v$
top $K \leq c-1$ eigenvectors $V_K = [v_1, v_2, \dots, v_K]$
3. Project X onto them $Z = X \cdot V_K$ & classify

$$1 \times K \quad 1 \times d \quad d \times K$$
4. $\hat{y}_c = \arg \max_c (z^T \Sigma^{-1} \mu - \frac{1}{2} \mu^T \Sigma^{-1} \mu + \log p(c))$ (vii)
Assuming each class c follows Gaussian with variance $\Sigma = N(Z; \mu_c, \Sigma)$

APPENDIX

APPENDIX

(i) symmetry of geodesic distance

$$d(C_1, C_2) = \|\log C C_1^{-1/2} C_2 C_1^{-1/2}\|_F$$

$$d(C_2, C_1) = \|\log C C_2^{-1/2} C_1 C_2^{-1/2}\|_F$$

$$\text{let } X = C_1^{-1/2} C_2 C_1^{-1/2}$$

$$(AB)^{-1} = B^{-1}A^{-1}$$

$$X^{-1} = (C_1^{-1/2} C_2 C_1^{-1/2})^{-1}$$

$$= (C_2 C_1^{-1/2})^{-1} (C_1^{-1/2})^{-1}$$

$$= (C_1^{-1/2})^{-1} C_2^{-1} (C_1^{-1/2})^{-1}$$

$$= C_1^{1/2} C_2^{-1} C_1^{1/2}$$

$$X^{-1} = C_1^{1/2} (C_2^{-1} C_1) C_1^{-1/2}$$

$$X = C_1^{1/2} (C_1^{-1} C_2) C_1^{-1/2}$$

$$\text{eig}(X^{-1}) = \text{eig}(C_2^{-1} C_1)$$

$$\text{eig}(X) = \text{eig}(C C_1^{-1} C_2)$$

Property: if $A = PAP^{-1}$, $\text{eig}(A) = \text{eig}(PAP^{-1})$

$$(C_2^{-1} C_1)^{-1} = C_1^{-1} C_2$$

$$\therefore \text{eig}(C C_2^{-1} C_1) = \text{reciprocal}(\text{eig}(C_1^{-1} C_2))$$

$$\|\log X\|_F = \left(\sum_i (\log \lambda_i)^2 \right)^{1/2}$$

$$\|\log X^{-1}\|_F = \left(\sum_i (\log \lambda_i^{-1})^2 \right)^{1/2} = \left(\sum_i (\log \lambda_i)^2 \right)^{1/2}$$

$$= \|\log X\|_F$$

$$\therefore \|\log X^{-1}\| = \|\log X\|_F \Rightarrow d(C_1, C_2) = d(C_2, C_1)$$

$$(ii) \text{Euclidean Mean} = \underset{C}{\text{argmin}} \sum_{i,a} d_{LE}^2(C, C_{i,a})$$

$$\nabla_C \sum_i d_{LE}^2(C, C_i) = 2 \sum_{i,a} (C - C_{i,a}) = 0$$

$$C_{E, \text{ref}} = \frac{1}{N_T} \sum_{i,a} C_{i,a}$$

$$(iii) \text{log-Euclidean Mean} = \log \left(\underset{C}{\text{argmin}} \sum_i d_{LE}^2(C, C_i) \right)$$

$$\nabla_C \sum_i d_{LE}^2(C, C_i) = 2 \sum_{i,a} (\log C - \log C_{i,a}) (C^{-1}) = 0$$

$$C_{LE, \text{ref}} = \log C = \frac{1}{N_T} \sum_{i,a} \log C_{i,a}$$

$$(iv) C_e^{-1} = C_{R, \text{ref}, i}^{-1/2} C_e C_{R, \text{ref}, i}^{-1/2}$$

Proof

$$C_{R, \text{ref}, i} = \text{SPD matrix. } C_{\text{ref}} = C_{\text{ref}}^T$$

We know $C_{R, \text{ref}, i}$ is unique for each i

So to generalize across a $i \in \{1, 2, 3, \dots, 1093\}$

$$C_{R, \text{ref}, 1} = C_{R, \text{ref}, 2} = \dots = C_{R, \text{ref}, 1093} = I$$

$$C_{R, \text{ref}, i} = I \text{ (All eigenvectors (axes) } \Rightarrow \text{unity)}$$

$$W C_{R, \text{ref}, i} W^T = I.$$

$$W = W^T = C_{R, \text{ref}, i}^{-1/2}$$

$$C_{R, \text{ref}, i}^{-1/2} C_{R, \text{ref}, i} C_{R, \text{ref}, i}^{-1/2} = I \text{ for all } i$$

$$C_{R, \text{ref}, i} = \underset{C}{\text{argmin}} \sum_e d_R^2(C, C_e)$$

$$= \underset{C}{\text{argmin}} \sum_e \|\log(C^{-1/2} C_e C^{-1/2})\|_F = C_{\text{ref}}$$

$$= \underset{C}{\text{argmin}} \sum_e \|\log(C^{1/2} C_{\text{ref}}^{-1/2} C_e C_{\text{ref}}^{-1/2} C^{-1/2})\|_F$$

$$= \underset{C}{\text{argmin}} \sum_e \|\log(C^{-1/2} C_{\text{ref}}^{-1/2} C_e C_{\text{ref}}^{-1/2} C^{-1/2})\|_F$$

$$= \underset{C}{\text{argmin}} \sum_e \|\log((C_{\text{ref}} C)^{-1/2} C_e (C_{\text{ref}} C)^{-1/2})\|_F$$

$$= \underset{C_{\text{ref}}}{\text{argmin}} \sum_e \|\log((C_{\text{ref}} C)^{-1/2} C_e (C_{\text{ref}} C)^{-1/2})\|_F$$

$$C_{\text{ref}}; C_{\text{ref}} = C_{\text{ref}} C \Rightarrow C_{\text{ref}} = I \text{ or } C = I$$

$\Rightarrow$ We know at this condition $C_{\text{ref}} = C_{\text{ref}}$.

$$\therefore C = I. \Rightarrow C_{\text{ref}} = I$$

We shaped all covariances such that we $C_{\text{ref}} = I$ (all eigenvectors are unit vectors)

(v) Signal whitening $\Rightarrow$ linear transformation

$$C_{i,a,e}^{-1} = C_{R, \text{ref}, i}^{-1/2} C_{i,a,e} C_{R, \text{ref}, i}^{-1/2}$$

$$= C_{R, \text{ref}, i}^{-1/2} \frac{1}{N-1} X_{i,a,e} X_{i,a,e}^T C_{R, \text{ref}, i}^{-1/2}$$

$$= \frac{1}{N-1} C_{R, \text{ref}, i}^{-1/2} X_{i,a,e} X_{i,a,e}^T C_{R, \text{ref}, i}^{-1/2}$$

$$= \frac{1}{N-1} (C_{R, \text{ref}, i}^{-1/2} X_{i,a,e}) (C_{R, \text{ref}, i}^{-1/2} X_{i,a,e})^T$$

$$= \frac{1}{N-1} X_{i,a,e}' X_{i,a,e}'^T$$

$$X_{i,a,e}' = C_{R, \text{ref}, i}^{-1/2} X_{i,a,e}$$

$$C \times C. C \times T \Rightarrow C \times T$$

$$\text{So } U_{\text{ref}} \Lambda_{\text{ref}}^{-1/2} U_{\text{ref}}^T X_{i,a,e}.$$

rotate back $\left[\begin{array}{c} \text{transform} \\ \text{rotate signal} \end{array} \right] \Rightarrow$ linear transform

We normalize the signal to the reference coordinates.

$$(vi) C_K \omega = \lambda (C_E) \omega$$

$$C_E^{-1/2} C_K C_E^{-1/2} \omega = \lambda (C_E^{-1/2} C_E C_E^{-1/2}) \omega$$

$$S_K \omega = \lambda \omega.$$

$$S_K = C_E^{-1/2} C_K C_E^{-1/2}$$

Let eigendecomposition of $C_E = E \Lambda E^T$

$$C_E^{-1/2} = E \Lambda^{-1/2} E^T.$$

eigendecomposition of $S_K = U \Lambda U^T$; $U^T U = I$

$$S_K = U \Lambda U^{-1} \Rightarrow U^{-1} S_K U = U^T S_K U = \Lambda$$

$$U^T C_E^{-1/2} C_K C_E^{-1/2} U = \Lambda \quad [C_E^{-1/2} = (C_E^{-1/2})^T]$$

$$(C_E^{-1/2} U)^T C_K (C_E^{-1/2} U) = \Lambda \text{ [diagonal]}.$$

$$W^T C_K W = \Lambda, \quad W^T C_E W = I.$$

$$W^T = W^{-1} C_E^{-1} \Rightarrow W^{-1} C_E^{-1} C_K W = \Lambda.$$

$$C_K W = \lambda C_E W \Rightarrow \text{this is verification}$$

$\therefore W = C_E^{-1/2} U$ where U is eigenvector of whitened covariance

(vii) Assume C classes, class c has Gaussian distribution. $N(x_c; \mu_c, \Sigma)$ with same Σ

Bayes theorem:

$$P(y = c/x) \propto P(x/y = c) P(y = c)$$

$$P(y = c) = \pi_c \text{ (prior)}$$

$$P(y = c/x) \propto P(x/y = c) \cdot \pi_c \text{ --- (1)}$$

$$x/y = c \sim N(\mu_c, \Sigma)$$

$$P(x/y = c) = \frac{1}{\sqrt{2\pi}\Sigma} e^{-\frac{(x-\mu_c)\Sigma^{-1}(x-\mu_c)^T}{2}}$$

From (1): $\log P(y = c/x) = \log (P(x/y = c)) + \log \pi_c$

$$\log (P(y = c/x)) = \log \left(\frac{1}{\sqrt{2\pi}\Sigma} e^{-\frac{(x-\mu_c)\Sigma^{-1}(x-\mu_c)^T}{2}} \right) + \log \pi_c$$

$$= -\frac{1}{2}(x-\mu_c)\Sigma^{-1}(x-\mu_c)^T - \frac{1}{2} \log 2\pi\Sigma + \log \pi_c$$

$$= -\frac{1}{2} x \Sigma^{-1} x^T + \frac{1}{2} x \Sigma^{-1} \mu_c^T + \frac{1}{2} \mu_c \Sigma^{-1} x^T - \frac{1}{2} \mu_c \Sigma^{-1} \mu_c^T$$

$$- \frac{1}{2} \log 2\pi\Sigma + \log \pi_c$$

$$= -\frac{1}{2} [x \Sigma^{-1} x - 2 x^T \Sigma^{-1} \mu_c + \mu_c \Sigma^{-1} \mu_c^T + \log 2\pi\Sigma] + \log \pi_c.$$

We need to maximize $P(y = c/x) \propto \log P(y = c/x)$

$\Rightarrow$ depends only on class c (changing factor)

$x \Sigma^{-1} x$, $\log 2\pi\Sigma$ are constant $\Rightarrow$ independent of c .

$$\text{maximize } \Rightarrow x^T \Sigma^{-1} \mu_c - \frac{1}{2} \mu_c \Sigma^{-1} \mu_c^T + \log \pi_c$$

$$c = \operatorname{argmax}_c \left(x^T \Sigma^{-1} \mu_c - \frac{1}{2} \mu_c \Sigma^{-1} \mu_c^T + \log \pi_c \right)$$

Case LDA: $x_k^T V_K = x_k^T$. Otherwise the $\mu_c \Rightarrow$ distribution of x_k^T . $N(x_k; \mu_{ck}, \Sigma_k)$

$$c = \operatorname{argmax}_c \left(x_k^T \Sigma_k^{-1} \mu_{ck} - \mu_{ck} \Sigma_k^{-1} \mu_{ck}^T + \log \pi_c \right)$$

